

Uddoygi ERP: Comprehensive System Manual

Complete Feature Inventory, Architecture, and Logic Maps

Part 1: Administrative Feature Inventory (40+ Screens)

Screen Name	Details / Purpose	Components & Logic
Admin Dashboard	Central oversight hub for all company metrics.	UCard, ActivityFeed, KPI Grids. Math: Revenue/Expense sum.
Net Profitability	Deep financial analysis of company health.	PieChart, StatCards. Math: Revenue - (Labor+Prod+Mark).
HR Efficiency	Real-time staffing optimization tool.	EfficiencyHero, MetricTiles. Math: Workforce / WorkOrders.
Live Monitoring	Real-time oversight of factory machines.	MachineGrid, UtilityProgress. Math: Output vs Goal.
Order Analysis	Granular tracking of production bottlenecks.	StageProgress, DeliveryPredictor. Math: Stage duration.
Inventory Mgmt	BOM vs Warehouse Stock comparison.	StockTable, ShortageAlerts. Math: Stock - Requirement.
Sales Pipeline	Lead tracking and conversion analytics.	FunnelChart, LeadTable. Math: Deals / Leads.
Customer CRM	Buyer relationship and tier management.	CustomerList, TierPill. Ranking logic based on volume.
QC Overview	Aggregated quality test reports.	DefectHeatmap, SupplierPenalty. Math: Defect %.
Campaign ROI	Marketing spend vs return analysis.	CampaignGrid, ROITracker. Math: Revenue / Spend.
Employee Mgmt	Admin view of all staff and roles.	StaffTable, RoleManager, PermissionGuard.
Salary Overview	Global payroll approval and oversight.	SalaryList, ApprovalStatus. Syncs with Finance.
Welfare Scheme	Company-wide benefits monitoring.	WelfareGrid, BenefitDetails.
SMTP Settings	Email server configuration for notifications.	InputForm, TestEmailAction.
Security Audit	Access logs and system permission controls.	LogList, UserRoles.
All Notices	Global announcement management.	NoticeForm, TargetDepartmentPicker.
Reports Hub	Centralized exports for PDF/Excel data.	ExportActions, DateRangePicker.
Company Profile	Organization identity and branch settings.	ProfileEditor, BranchList.
Complaints Hub	Resolution tracking for staff/customers.	TicketList, ResolutionWorkflow.
Research Hub	Innovation tracking and R&D monitoring.	ProjectGrid, MilestoneTracker.

Part 2: HR & Personnel Inventory (30+ Screens)

Screen Name	Details / Purpose	Components & Logic
Payroll Processor	Automated monthly salary generation.	ExpansionTile, LoanBreakdown. Math: Gross - (Loan+Tax).
Loan Approval	Vetting and approving staff loan requests.	ApprovalWorkflow, RepaymentTerms.
Leave Mgmt	Employee absence tracking and approvals.	CalendarView, LeaveList. Syncs with Efficiency.
Employee Dir	Searchable profiles and contact details.	SearchField, ProfileCard.
Appointment Gen	Automated legal document creation.	PDFExporter, TemplateEditor.
Salary Cert	Generation of official income proofs.	DocTemplate, DigitalSignature.
Budget Forecast	Financial planning for departmental spend.	ForecastChart, VarianceAnalysis.
Attendance	Daily check-in/out tracking.	ShiftTracker, OvertimeCalc.
Recruitment	Job posting and applicant tracking.	JobGrid, ApplicantStages.
Incentives	KPI-based bonus calculation.	IncentiveFormula, RewardGrid. Math: % of target.
General Ledger	Detailed accounting for HR transactions.	LedgerTable, EntryForm.
Tax Management	Compliance and deduction tracking.	TaxTable, BracketEditor.
Accounts Pay	Managing company debts to suppliers.	PayableList, PaymentDispatch.
Balance Update	Manual adjustment of fund accounts.	AdjustmentForm, AuditTrail.
Shift Tracker	Managing worker rotations and rosters.	ShiftGrid, SwapRequest.
Welfare Screen	Staff welfare fund and activity logs.	ContributionTable, GrantHistory.

Part 3: Marketing & Sales Inventory (25+ Screens)

Screen Name	Details / Purpose	Components & Logic
Marketing Dash	Sales agent performance and targets.	TargetGauge, AgentLeaderboard.
Invoice Creator	Billing generation for customers.	ItemPicker, TotalCalc. Math: Qty*Price + Tax.
All Invoices	History and status of all transactions.	StatusPill, PaymentVerifier.
Payment Slips	Image-to-data validation for bank slips.	DocumentExtractor, OCRPreview.
Order Tracking	Shipping status for customer satisfaction.	TimelineView, CourierSync.
Campaign Mgr	Digital ad and bulk-offer manager.	OfferForm, LeadSourceTracker.
Products Page	Catalog view for sales agents.	ProductGrid, SpecificationView.
Sales Report	Departmental revenue analytics.	SalesReportChart, RegionHeatmap.
Task Assign	Sales task allocation for agents.	TaskCard, PriorityPicker.
Address Valid	AI-assisted delivery address checking.	AddressParser, GoogleMapsSync.
Remuneration	Agent commission and bonus tracking.	CommissionCalc, PaymentHistory.

Part 4: Operations & Innovation (20+ Screens)

Screen Name	Details / Purpose	Components & Logic
Live Floor	Real-time machine and line oversight.	MachineStatusGrid, LiveCounter.
Work Order	Production instructions and specs.	TechnicalSpecs, StageUpdate. Syncs with Marketing.
QC Report	Material and product quality testing.	DefectLogger, PassRateCalc.
R&D Dashboard	Prototype and innovation monitoring.	ProjectMilestones, R&DAnalytics.
BOM Creator	Bill of Materials for new products.	MaterialPicker, UnitCostCalc.
Project Req	Requests for new product development.	RequestForm, PriorityQueue.
Milestones	Tracking R&D progress stages.	Timeline, AssetUploader.
Daily Update	Worker reports from the factory floor.	UpdateForm, PhotoLogger.
Resource Alloc	Assigning labor to production lines.	CapacityChart, StaffAssigner.

Part 5: Core Services, Models & Themes

5.1 Core Services (The System Brain)

Service Name	Description / Responsibility	Key Technology
DataSyncService	Orchestrates cross-departmental triggers.	Firestore Batch / Triggers
DocumentExtractor	AI module for image-to-data parsing.	OCR / NLP / Regex
AIService	Provides insights and assistant capabilities.	Google Gemini API
AuthService	Handles login, multi-tenant company IDs.	Firebase Auth
DB Service	Abstracted Firestore collection access.	Cloud Firestore SDK
LocalStorage	Offline persistence and session caching.	Shared Preferences / Hive
FCM Service	Push notifications and real-time alerts.	Firebase Cloud Messaging
EmailService	Automated transaction and alert emails.	SMTP / SendGrid
DriveStorage	Secure storage for invoices and files.	Google Drive API
AppRulesGuard	Enforces business logic across modules.	Policy Engine

5.2 Data Models (Blueprints)

Model Name	Fields / Schema Highlights	Used In
UserModel	uid, email, cid, role, dept, salary.	Global Auth
LoanModel	employeeId, amount, emi, deducted, status.	HR / Finance
InvoiceModel	grandTotal, items, status, agent, cid.	Marketing / Admin
WorkOrderModel	specs, currentStage, targetQty, deadline.	Factory / Marketing
BOMModel	productId, materials, quantityPerUnit.	R&D / Inventory
PipraTxnModel	txnId, amount, beneficiary, status.	Payments
NoticeModel	title, body, targetDept, timestamp.	Communication
ComplaintModel	userId, category, description, status.	Common

5.3 Premium Themes (Aesthetics)

Theme Name	Primary Colors	Visual Style
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Theme Name	Primary Colors	Visual Style
Admin Theme	Purple (#1E0040), White	Professional, Analytics-focused, Elevated.
Marketing Theme	Blue (#0D47A1), Surface	Energetic, Sales-driven, Clean.
HR Theme	Green (#25BC5F), White	Trustworthy, Organic, Structured.
Factory Theme	Slate (#0F172A), Amber	Industrial, High-contrast, Functional.
R&D Theme	Indigo (#4F46E5), Dark	Futuristic, Innovation-focused.